

Tutorial-3

A retail Company wants to predict Monthly Sales revenue based on Marketing & Operational factors.

Input Features:

- Advertising Spend (£ in thousands)
- No. of Sales persons
- Store footfall (per Month)
- Discount rate (%)

Ad spend.	Salesperson	Footfall	Discount	Sales revenue
50	5	1200	10	8.5
70	6	1500	12	10.2
40	4	1000	8	6.8
90	8	1900	15	14.0
60	6	1400	10	9.5
80	7	1700	14	12.8
55	5	1300	9	8.9
100	9	2100	18	16.0
65	6	1450	4	9.8
45	4	1100	8	7.2

1. Define Dependent Variables:

Dependent Variables is Monthly Sales revenue because MSR depends on other factors like Advertising Spend number of Sales persons, Store football, & Discount type.

2. Write the Multiple Linear Regression Model.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

where Y = Sales Revenue

X_1 = Advertising Spend

X_2 = No. of Sales person

X_3 = Store football

X_4 = Discount rate

β_0 = Intercept

$\beta_1, \beta_2, \beta_3, \beta_4$ = Co. coefficients.

3. Final Model.

$$\text{Sales} = \beta_0 + \beta_1 (\text{Ad Spend}) + \beta_2 (\text{Sales persons}) + \beta_3 (\text{Football}) + \beta_4 (\text{Discount})$$

Substitute each row.

$$\text{Row 1: } 25 = \beta_0 + 50\beta_1 + 15\beta_2 + 1200\beta_3 + 10\beta_4$$

$$\text{Row 2: } 10.2 = \beta_0 + 10\beta_1 + 4\beta_2 + 1500\beta_3 + 12\beta_4$$

$$\text{Row 3: } 6.8 = \beta_0 + 40\beta_1 + 4\beta_2 + 1000\beta_3 + 8\beta_4$$

$$\text{Row 4: } 14 = \beta_0 + 90\beta_1 + 8\beta_2 + 1900\beta_3 + 15\beta_4$$

$$\text{Row 5: } 7.5 = \beta_0 + 60\beta_1 + 6\beta_2 + 1400\beta_3 + 10\beta_4$$

$$\text{Row 6: } 12.5 = \beta_0 + 80\beta_1 + 7\beta_2 + 1700\beta_3 + 24\beta_4$$

$$\text{Row 7: } 89 = \beta_0 + 55\beta_1 + 5\beta_2 + 1300\beta_3 + 9\beta_4$$

$$\text{Row 8: } 16.0 = \beta_0 + 100\beta_1 + 9\beta_2 + 2100\beta_3 + 12\beta_4$$

$$\text{Row 9: } 9.8 = \beta_0 + 65\beta_1 + 6\beta_2 + 1400\beta_3 + 11\beta_4$$

$$\text{Row 10: } 7.2 = \beta_0 + 45\beta_1 + 4\beta_2 + 1100\beta_3 + 8\beta_4$$

Discuss the effect of Discount rate on Sales growth revenue

Discount = 8, Sales = 6.8, Discount = 18, Sales = 16,

Since $1828 \Rightarrow 16 > 6.8$

4. Find MSE for 5 iterations:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$\hat{y}_i = [8, 9, 7, 13, 10, 11, 9, 15, 10, 8]$$

Sq. Errors:

$$(0.5)^2 + (1.2)^2 + (-0.2)^2 + (1)^2 + (-0.5)^2 + (-0.1)^2 + (1)^2 + (-0.2)^2 + (-0.8)^2$$

$$= 6.92$$

$$MSE = \frac{6.92}{10} \Rightarrow 0.692$$

Iteration 2:

$$\hat{Y}_2 = [-8.3, 9.8, 6.9, 13.6, 9.7, 12, 8.6, 15.5, 9.9, 7.5]$$

Sum of Squares: 201

$$MSE = \frac{201}{10} = 0.201$$

Iteration 3:

$$\hat{Y}_3 = [8.4, 10, 6.85, 13.8, 9.65, 12.4, 8.88, 15.9, 9.82, 7.28]$$

$$SSE = 0.18$$

$$SSE = 0.62$$

$$MSE = \frac{0.18}{10} = 0.018$$

$$MSE = \frac{0.62}{10} = 0.062$$

Iteration 4:

$$\hat{Y} = [8.45, 10.1, 6.82, 13.9, 9.55, 12.4, 8.88, 15.9, 9.82, 7.28]$$

$$SSE = 0.18$$

$$MSE = \frac{0.18}{10} = 0.018$$

Iteration 5:

$$\hat{Y} = [8.5, 10.21, 6.8, 14, 9.5, 12.5, 8.9, 15.9, 9.82, 7.29]$$

$$SSE = 0$$

$$MSE = \frac{0}{10} = 0$$